

Object Oriented Design Quality Metrics

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.3	org.axonframework.messaging.core	core	12499	39679	
1	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	11625	38453	
2	axon-common-5.0.3	org.axonframework.common.configuration	configuration	3727	19786	
3	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	3012	10890	
4	axon-common-5.0.3	org.axonframework.common.annotation	annotation	2579	2787	
5	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	1497	5379	
6	axon-common-5.0.3	org.axonframework.common.infra	infra	1378	1669	
7	axon-common-5.0.3	org.axonframework.common	common	977	2407	
8	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	688	2544	
9	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	annotation	512	1632	
10	axon-conversion-5.0.3	org.axonframework.conversion	conversion	468	1807	
11	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	425	2147	
12	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	293	833	
13	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore	eventstore	224	724	
14	axon-messaging-5.0.3	org.axonframework.messaging.tracing	tracing	136	478	
15	axon-test-5.0.3	org.axonframework.test.fixture	fixture	134	376	
16	axon-modelling-5.0.3	org.axonframework.modelling	modelling	125	282	
17	axon-modelling-5.0.3	org.axonframework.modelling.entity	entity	115	394	
18	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	child	89	230	
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	segmenting	86	662	

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.3	org.axonframework.messaging.core	core	18180	59941	
1	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	11275	37337	
2	axon-common-5.0.3	org.axonframework.common	common	8038	25190	
3	axon-common-5.0.3	org.axonframework.common.configuration	configuration	3566	19095	
4	axon-common-5.0.3	org.axonframework.common.annotation	annotation	2578	2786	
5	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	2061	7468	
6	axon-common-5.0.3	org.axonframework.common.infra	infra	1364	1595	
7	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	900	3463	
8	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	annotation	423	1325	
9	axon-conversion-5.0.3	org.axonframework.conversion	conversion	397	1597	
10	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	350	1416	
11	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	269	973	
12	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	241	688	
13	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	processing	218	685	
14	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	streaming	207	655	
15	axon-messaging-5.0.3	org.axonframework.messaging.tracing	tracing	107	356	
16	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore	eventstore	76	207	
17	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	segmenting	74	623	
18	axon-modelling-5.0.3	org.axonframework.modelling	modelling	74	226	
19	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	69	225	

Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the code unexpectedly ("fragile code"). As soon as more effort is required, keeping up becomes

difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoi
0	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	configuration	2694	16653	
1	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	pooled	2322	9835	
2	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	event	1282	4029	
3	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore.jpa	jpa	1239	4839	
4	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.configuration	configuration	1237	4808	
5	axon-messaging-5.0.3	org.axonframework.messaging.core.interception	interception	1160	6002	
6	axon-test-5.0.3	org.axonframework.test.fixture	fixture	1079	3869	
7	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	annotation	857	3255	
8	axon-modelling-5.0.3	org.axonframework.modelling.entity	entity	800	2887	
9	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	subscribing	750	2762	
10	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore	eventstore	716	2684	
11	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.autoconfig	autoconfig	714	2921	
12	axon-modelling-5.0.3	org.axonframework.modelling.configuration	configuration	585	2279	
13	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	533	2104	
14	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.dist...	distributed	513	1412	
15	axon-messaging-5.0.3	org.axonframework.messaging.monitoring.interce...	interception	500	1550	
16	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.in...	interception	499	1240	
17	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.inte...	interception	491	1561	
18	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.anno...	annotation	480	1539	
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	471	1201	

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgo
0	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	675	2160	
1	axon-messaging-5.0.3	org.axonframework.messaging.core	core	434	1531	
2	axon-modelling-5.0.3	org.axonframework.modelling	modelling	372	1088	
3	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	processing	348	1280	
4	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	325	1035	
5	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	316	1128	
6	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing	eventsourcing	313	1037	
7	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	streaming	275	1008	
8	axon-modelling-5.0.3	org.axonframework.modelling.entity	entity	269	854	
9	axon-server-connector-5.0.3	org.axonframework.axonserver.connector	connector	253	866	
10	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	pooled	231	927	
11	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	188	716	
12	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.eventstore	eventstore	178	572	
13	axon-modelling-5.0.3	org.axonframework.modelling.entity.annotation	annotation	178	679	
14	axon-messaging-5.0.3	org.axonframework.messaging.core.interception	interception	172	588	
15	axon-test-5.0.3	org.axonframework.test	test	169	510	
16	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	event	142	416	
17	axon-test-5.0.3	org.axonframework.test.fixture	fixture	138	490	
18	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.autoconfig	autoconfig	129	452	
19	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	annotation	128	341	

Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

Instability is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.3	org.axonframework.common.annotation	annotation	0.001162	0.020408	0.000000	
1	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	0.009205	0.079848	0.142857	
2	axon-common-5.0.3	org.axonframework.common.infra	infra	0.019217	0.075758	0.090909	
3	axon-messaging-5.0.3	org.axonframework.messaging.core	core	0.026558	0.160321	0.170000	
4	axon-common-5.0.3	org.axonframework.common	common	0.030754	0.055738	0.000000	
5	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	0.042793	0.134328	0.333333	
6	axon-conversion-5.0.3	org.axonframework.conversion	conversion	0.042945	0.138889	0.250000	
7	axon-common-5.0.3	org.axonframework.common.configuration	configuration	0.070574	0.256158	0.283784	
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	0.135228	0.219895	0.400000	
9	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	0.165552	0.259259	0.431818	
10	axon-common-5.0.3	org.axonframework.common.util	util	0.200000	0.153846	0.000000	
11	axon-messaging-5.0.3	org.axonframework.messaging.monitoring	monitoring	0.213592	0.190476	0.571429	
12	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	0.230769	0.178571	0.571429	
13	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	store	0.263158	0.190476	1.000000	
14	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	0.263819	0.320513	0.461538	
15	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	0.333333	0.444444	0.000000	
16	axon-messaging-5.0.3	org.axonframework.messaging.tracing	tracing	0.333333	0.320755	0.533333	
17	axon-common-5.0.3	org.axonframework.common.tx	tx	0.375000	0.230769	0.666667	
18	axon-update-5.0.3	org.axonframework.update.api	api	0.379310	0.384615	0.000000	
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.380952	0.384615	0.750000	

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.3	org.axonframework.common.annotation	annotation	0.000775	0.013793	0.000000	
1	axon-common-5.0.3	org.axonframework.common.infra	infra	0.003652	0.025000	0.000000	
2	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	0.016401	0.061728	0.500000	
3	axon-messaging-5.0.3	org.axonframework.messaging.core	core	0.023316	0.200436	0.397906	
4	axon-common-5.0.3	org.axonframework.common.configuration	configuration	0.030188	0.067227	0.436170	
5	axon-conversion-5.0.3	org.axonframework.conversion	conversion	0.059242	0.136364	0.500000	
6	axon-common-5.0.3	org.axonframework.common.util	util	0.090909	0.100000	0.000000	
7	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	0.126582	0.153846	0.500000	
8	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	0.154589	0.318182	0.500000	
9	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	0.157343	0.204082	0.500000	
10	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	0.166667	0.250000	0.000000	
11	axon-modelling-5.0.3	org.axonframework.modelling.repository	repository	0.204819	0.263158	0.500000	
12	axon-messaging-5.0.3	org.axonframework.messaging.tracing	tracing	0.207407	0.300000	0.461538	
13	axon-common-5.0.3	org.axonframework.common.tx	tx	0.230769	0.230769	0.500000	
14	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.230769	0.333333	0.500000	
15	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	annotation	0.232305	0.316667	0.500000	
16	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	0.246711	0.514706	0.567901	
17	axon-common-5.0.3	org.axonframework.common.property	property	0.250000	0.400000	0.000000	
18	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.util	util	0.250000	0.333333	0.000000	
19	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	0.265306	0.544118	0.527273	

Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
0	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	41
1	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot	springboot	0.0	0	19
2	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	event	0.0	0	14
3	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	11
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8
5	axon-messaging-5.0.3	org.axonframework.messaging.core.timeout	timeout	0.0	0	8
6	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	6
7	axon-messaging-5.0.3	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6
8	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	command	0.0	0	6
9	axon-update-5.0.3	org.axonframework.update.api	api	0.0	0	6
10	axon-conversion-5.0.3	org.axonframework.conversion.jackson	jackson	0.0	0	5
11	axon-conversion-5.0.3	org.axonframework.conversion.converter	converter	0.0	0	5
12	axon-conversion-5.0.3	org.axonframework.conversion.jackson2	jackson2	0.0	0	5
13	axon-messaging-5.0.3	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5
14	axon-tracing-opentelemetry-5.0.3	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5
15	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	0.0	0	5
16	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4
17	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer...	springboot	0.0	0	4
18	axon-update-5.0.3	org.axonframework.update.detection	detection	0.0	0	4
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3
20	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3
21	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3
22	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2
23	axon-messaging-5.0.3	org.axonframework.messaging.core.reflection	reflection	0.0	0	2
24	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2
25	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jdbc	0.0	0	2
26	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jpa	0.0	0	2
27	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	configuration	0.0	0	2
28	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2
29	axon-test-5.0.3	org.axonframework.test.server	server	0.0	0	2

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
146	axon-common-5.0.3	org.axonframework.common.function	function	1.000000	2	2
147	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	streaming	1.000000	1	1
148	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuator	actuator	1.000000	1	1
149	axon-common-5.0.3	org.axonframework.common.tx	tx	1.000000	1	1
144	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4
145	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4
143	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14
142	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation	annotation	0.700000	7	10
141	axon-modelling-5.0.3	org.axonframework.modelling.repository	repository	0.625000	5	8
140	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	transaction	0.600000	3	5
138	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7
139	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.util	util	0.571429	4	7
137	axon-modelling-5.0.3	org.axonframework.modelling.configuration	configuration	0.538462	7	13
131	axon-common-5.0.3	org.axonframework.common.jdbc	jdbc	0.500000	6	12
132	axon-test-5.0.3	org.axonframework.test.extension	extension	0.500000	2	4
133	axon-common-5.0.3	org.axonframework.common.annotation	annotation	0.500000	2	4
134	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2
135	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2
136	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2
130	axon-common-5.0.3	org.axonframework.common.configuration	configuration	0.478261	22	46
128	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	0.473684	9	19
129	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	0.473684	9	19
127	axon-messaging-5.0.3	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11
125	axon-test-5.0.3	org.axonframework.test.fixture	fixture	0.437500	14	32
126	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16
123	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	child	0.428571	6	14
124	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing	eventsourcing	0.428571	3	7
122	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	0.416667	10	24
119	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10
120	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10

Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*
- Set the property "abstractnessIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
0	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	16	
1	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration	configuration	0.0	0	5	
2	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	41	
3	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.event	event	0.0	0	14	
4	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8	
5	axon-messaging-5.0.3	org.axonframework.messaging.core.timeout	timeout	0.0	0	8	
6	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	6	
7	axon-messaging-5.0.3	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6	
8	axon-server-connector-5.0.3	org.axonframework.axonserver.connector.command	command	0.0	0	6	
9	axon-update-5.0.3	org.axonframework.update.api	api	0.0	0	6	
10	axon-conversion-5.0.3	org.axonframework.conversion.jackson	jackson	0.0	0	5	
11	axon-conversion-5.0.3	org.axonframework.conversion.converter	converter	0.0	0	5	
12	axon-conversion-5.0.3	org.axonframework.conversion.jackson2	jackson2	0.0	0	5	
13	axon-messaging-5.0.3	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5	
14	axon-tracing-opentelemetry-5.0.3	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5	
15	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	0.0	0	5	
16	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4	
17	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer...	springboot	0.0	0	4	
18	axon-update-5.0.3	org.axonframework.update.detection	detection	0.0	0	4	
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3	
20	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3	
21	axon-messaging-5.0.3	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3	
22	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2	
23	axon-messaging-5.0.3	org.axonframework.messaging.core.reflection	reflection	0.0	0	2	
24	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2	
25	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jdbc	0.0	0	2	
26	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork.tr...	jpa	0.0	0	2	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
27	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2	
28	axon-test-5.0.3	org.axonframework.test.server	server	0.0	0	2	
29	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.retry	retry	0.0	0	1	

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	max5
119	axon-common-5.0.3	org.axonframework.common.function	function	1.000000	2	2	
120	axon-common-5.0.3	org.axonframework.common.tx	tx	1.000000	1	1	
117	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4	
118	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4	
116	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14	
115	axon-modelling-5.0.3	org.axonframework.modelling.repository	repository	0.625000	5	8	
114	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7	
113	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation	annotation	0.562500	9	16	
112	axon-modelling-5.0.3	org.axonframework.modelling.configuration	configuration	0.538462	7	13	
105	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.util	util	0.500000	4	8	
106	axon-common-5.0.3	org.axonframework.common.jdbc	jdbc	0.500000	6	12	
107	axon-test-5.0.3	org.axonframework.test.extension	extension	0.500000	2	4	
108	axon-common-5.0.3	org.axonframework.common.annotation	annotation	0.500000	2	4	
109	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2	
110	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2	
111	axon-messaging-5.0.3	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2	
104	axon-common-5.0.3	org.axonframework.common.configuration	configuration	0.478261	22	46	
103	axon-messaging-5.0.3	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11	
101	axon-test-5.0.3	org.axonframework.test.fixture	fixture	0.437500	14	32	
102	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16	
100	axon-modelling-5.0.3	org.axonframework.modelling.entity.child	child	0.428571	6	14	
97	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10	
98	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10	
99	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.gateway	gateway	0.400000	2	5	
96	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling	queryhandling	0.380952	24	63	
95	axon-modelling-5.0.3	org.axonframework.modelling	modelling	0.380435	35	92	
94	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.ga...	gateway	0.375000	3	8	
93	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.dist...	distributed	0.363636	4	11	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxS
92	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	0.359375	23	64	
90	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.replay	replay	0.357143	5	14	

Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-5.0.3	org.axonframework.messaging.tracing.attributes	attributes	NaN	0.0	NaN	6
1	axon-conversion-5.0.3	org.axonframework.conversion.converter	converter	NaN	0.0	NaN	5
2	axon-messaging-5.0.3	org.axonframework.messaging.core.reflection	reflection	NaN	0.0	NaN	2
3	axon-common-5.0.3	org.axonframework.common.function	function	NaN	1.0	NaN	2
4	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.retry	retry	NaN	0.0	NaN	1
5	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuator	actuator	NaN	1.0	NaN	1
6	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics.micrometer...	reservoir	NaN	0.0	NaN	1
7	axon-common-5.0.3	org.axonframework.common.io	io	NaN	0.0	NaN	1
8	axon-common-5.0.3	org.axonframework.common.digest	digest	NaN	0.0	NaN	1
9	axon-conversion-5.0.3	org	org	NaN	0.0	NaN	0
10	axon-conversion-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
11	axon-messaging-5.0.3	org	org	NaN	0.0	NaN	0
12	axon-messaging-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
13	axon-messaging-5.0.3	org.axonframework.messaging	messaging	NaN	0.0	NaN	0
14	axon-spring-boot-autoconfigure-5.0.3	org	org	NaN	0.0	NaN	0
15	axon-spring-boot-autoconfigure-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
16	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension	extension	NaN	0.0	NaN	0
17	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.service	service	NaN	0.0	NaN	0
18	axon-metrics-micrometer-5.0.3	org	org	NaN	0.0	NaN	0
19	axon-metrics-micrometer-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
20	axon-metrics-micrometer-5.0.3	org.axonframework.extension	extension	NaN	0.0	NaN	0
21	axon-metrics-micrometer-5.0.3	org.axonframework.extension.metrics	metrics	NaN	0.0	NaN	0
22	axon-test-5.0.3	org	org	NaN	0.0	NaN	0
23	axon-test-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
24	axon-eventsourcing-5.0.3	org	org	NaN	0.0	NaN	0
25	axon-eventsourcing-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
26	axon-server-connector-5.0.3	org	org	NaN	0.0	NaN	0
27	axon-server-connector-5.0.3	org.axonframework	axonframework	NaN	0.0	NaN	0
28	axon-server-connector-5.0.3	org.axonframework.axonserver	axonserver	NaN	0.0	NaN	0
29	axon-tracing-opentelemetry-5.0.3	org	org	NaN	0.0	NaN	0

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-common-5.0.3	org.axonframework.common.lifecycle	lifecycle	0.833333	0.000000	0.166667	5
1	axon-conversion-5.0.3	org.axonframework.conversion	conversion	0.797901	0.142857	0.059242	35
2	axon-common-5.0.3	org.axonframework.common.infra	infra	0.746348	0.250000	0.003652	8
3	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	token	0.716378	0.129032	0.154589	31
4	axon-messaging-5.0.3	org.axonframework.messaging.core	core	0.697023	0.279661	0.023316	236
5	axon-update-5.0.3	org.axonframework.update.api	api	0.684211	0.000000	0.315789	6
6	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling.co...	configuration	0.678571	0.750000	0.928571	4
7	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.conf...	configuration	0.678571	0.750000	0.928571	4
8	axon-messaging-5.0.3	org.axonframework.messaging.core.unitofwork	unitofwork	0.675907	0.307692	0.016401	26
9	axon-test-5.0.3	org.axonframework.test.server	server	0.666667	0.000000	0.333333	2
10	axon-common-5.0.3	org.axonframework.common.util	util	0.575758	0.333333	0.090909	6
11	axon-messaging-5.0.3	org.axonframework.messaging.tracing	tracing	0.529435	0.263158	0.207407	19
12	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.anno...	annotation	0.523810	0.571429	0.952381	7
13	axon-update-5.0.3	org.axonframework.update.detection	detection	0.500000	0.000000	0.500000	4
14	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.util.jpa	jpa	0.500000	0.000000	0.500000	1
15	axon-update-5.0.3	org.axonframework.update.common	common	0.500000	0.000000	0.500000	1
16	axon-common-5.0.3	org.axonframework.common.annotation	annotation	0.499225	0.500000	0.000775	4
17	axon-common-5.0.3	org.axonframework.common.configuration	configuration	0.491551	0.478261	0.030188	46
18	axon-eventsourcing-5.0.3	org.axonframework.eventsourcing.annotation	annotation	0.474781	0.562500	0.912281	16
19	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.435897	0.333333	0.230769	3
20	axon-messaging-5.0.3	org.axonframework.messaging.core.annotation	annotation	0.434362	0.333333	0.232305	51
21	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling	eventhandling	0.430308	0.322981	0.246711	161
22	axon-test-5.0.3	org.axonframework.test.fixture	fixture	0.423214	0.437500	0.985714	32
23	axon-common-5.0.3	org.axonframework.common.property	property	0.416667	0.333333	0.250000	9
24	axon-modelling-5.0.3	org.axonframework.modelling.configuration	configuration	0.415174	0.538462	0.876712	13
25	axon-messaging-5.0.3	org.axonframework.messaging.eventstreaming	eventstreaming	0.405157	0.437500	0.157343	16
26	axon-spring-boot-autoconfigure-5.0.3	org.axonframework.extension.springboot.actuato...	axonserver	0.400000	0.000000	0.600000	2
27	axon-messaging-5.0.3	org.axonframework.messaging.queryhandling.gateway	gateway	0.400000	0.500000	0.900000	2
28	axon-messaging-5.0.3	org.axonframework.messaging.eventhandling.conf...	configuration	0.388083	0.714286	0.673797	14
29	axon-messaging-5.0.3	org.axonframework.messaging.commandhandling	commandhandling	0.375319	0.359375	0.265306	64

Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages
- Draw the "main sequence" as dashed green diagonal line

- Scale the packages by the number of types they contain
- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

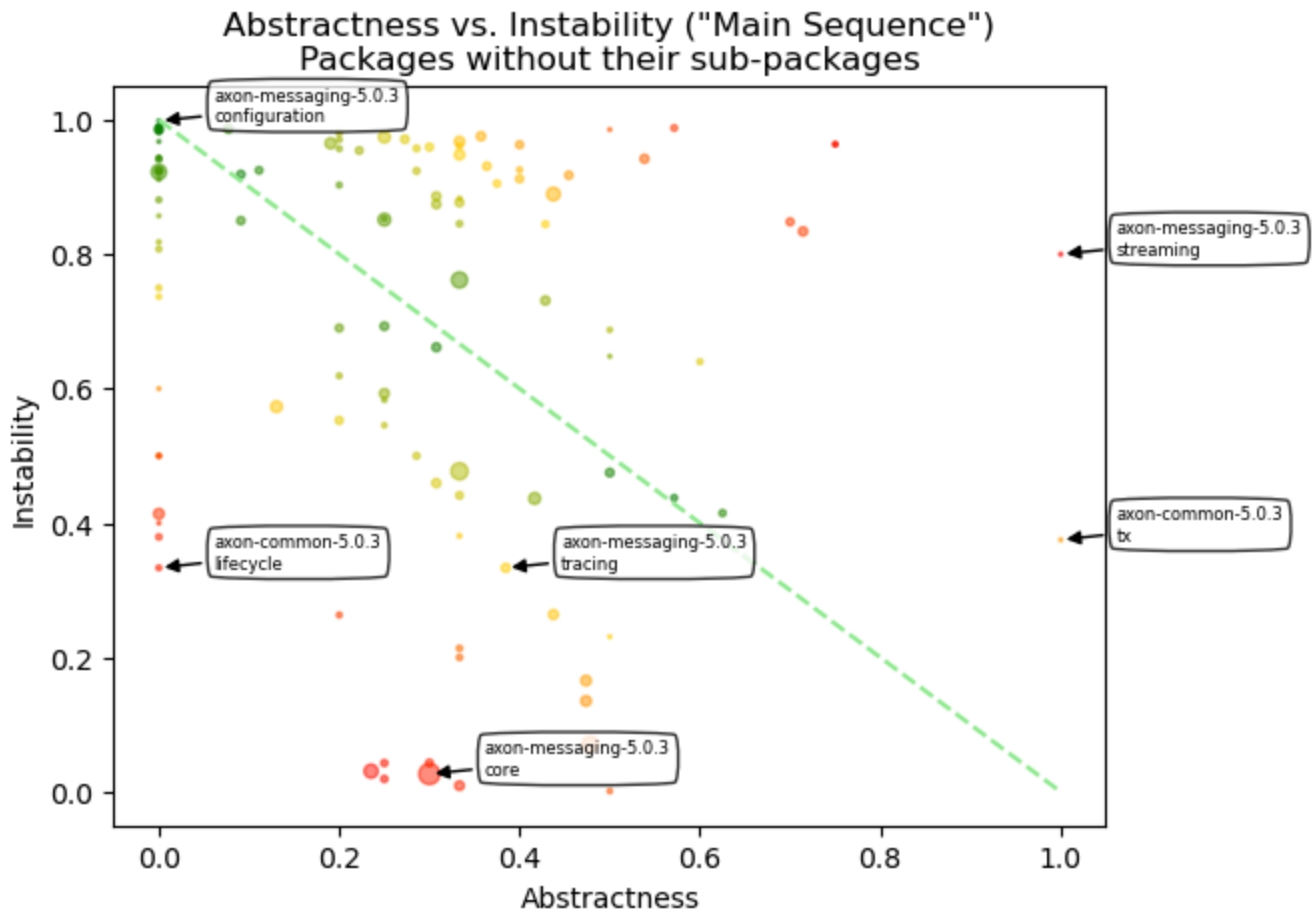


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")
Packages including their sub-packages

